



Fox ESS EV Charger

Modbus TCP Protocol 1.6

FOX ESS CO.,LTD

File number	MTWH-2023-0001	Version	V1.6
Documentation	SOFT.GRP	Auditor	Chen

Date: August 12, 2024

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Attention:

The register address was changed, and some contents were also modified.

1 Protocol Description

1.1 Acronyms Description

- **EMS:** Energy Management System
- **EVC:** Electric Vehicle Charger
- **INV:** Inverter
- **PV:** Photovoltaic
- **BAT:** Battery
- **SMT:** Smart Meter
- **mDNS:** Multicast Domain Name System

1.2 Target Group

This protocol is intended only for use by persons with technical knowledge and programming skills in the relevant field. If you have any questions, please contact our technical support staff. Please note that this protocol must be used with the permission of the Company. We do not assume any responsibility for any adverse consequences caused by private use.

1.3 Application Scenarios

1.3.1 Scenario Overview

Green and environmentally friendly charging scenarios are becoming more

and more popular, such as realizing green energy closed loop, improving clean energy utilization efficiency, reducing carbon emissions, and reducing charging costs. Through EMS, photovoltaic systems, smart loads and multiple EVC are connected together through communication networks to achieve centralized equipment management and intelligent energy deployment. At the same time, it is equipped with a cloud monitoring system to realize remote data monitoring. This efficient and transparent way of using energy has benefited more and more families.

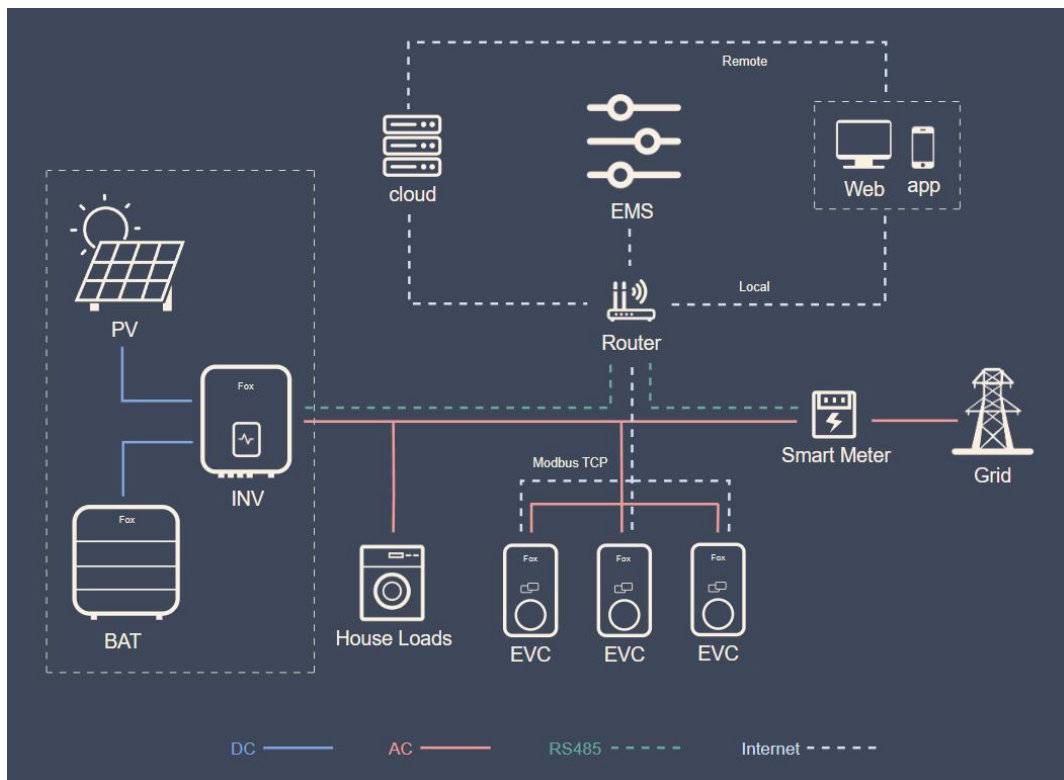


Figure 1. Usage scenario diagram

In this scenario, EMS manages the energy use of EVC, which is an important step in achieving efficient energy utilization. An efficient way to manage energy is to connect the EMS and EVC to the same network through a router. Then, after the EMS establishes communication with EVC through the Modbus TCP

Protocol, it controls one or more EVC in a reasonable way. This includes starting charging at the right time, controlling the charging power of EVC, and viewing the current charging data of EVC locally or remotely.

1.3.2 Discovery and Connection of EVC

The EMS communicates with the EVC through the local network port. An important part of establishing communication is that the EMS obtains the IP and port of the EVC. Here we use mDNS technology. The EVC will broadcast its master name and local IP, the EMS gets the IP of the EVC after obtaining the broadcast command, and then actively establishes a TCP/IP connection with the EVC. Finally, using Modbus TCP protocol, a complete communication is established.

The specific process is as follows:

(1) The EVC accesses the LAN through Ethernet or WIFI, and automatically obtains IP through DHCP;

(2) EVC sends information about the mDNS service whose multicast type is `_modbus._tcp.local` and instance name is `EVC-<SN>._modbus._tcp.local`, and SN indicates the serial number of EVC. For example, `EVC-60AE74B037KQ012._modbus._tcp.local`;

(3) The main control device on the same LAN uses the mDNS service to scan the instance with the type `_modbus._tcp.local` and obtains the IP and port of EVC from the instance;

(4) The master device is connected to the Modbus TCP service of the EVC by IP and port.

_modbus._tcp.local	EVC-60AE74B837KQ013._modbus._tcp.local	EVC-60AE74B837KQ013.local	192.168.9.51,192.168.9.77	502	ServiceInfo(type='_modbus._tcp', name='EVC-60AE74B837KQ013._modbus._tcp.local', addresses=[('192.168.9.51', 502), ('192.168.9.77', 502)], weight=0, priority=0, server='EVC-60AE74B837KQ013.local', properties={'b': None}, interface_index=None)
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1.4 Protocol Instructions

(1) The applicable firmware versions are A022KS1-E-2 series 1.14 or later, A011KS1-E-2 series 1.14 or later, and A7300S1-E-2 series 1.06 or later;

(2) The device address in the protocol can be configured using the company's EVC management APP "Fox Switch". For detailed operations, please view the instruction manual.

(3) The charging station must be connected to the same network (LAN) as the application;

(4) The default port of the server is 502;

(5) Unless otherwise specified, the sending interval of each message is 500 ms.

1.5 Protocol Overview

Modbus TCP Protocol is a communication protocol based on TCP/IP network, which allows data exchange between a master and several slaves. This protocol realizes the control of the master from the slave and the transmission

of data from the slave to the master. When communicating using the Modbus TCP Protocol, each participant must have a unique address, with address 0 reserved for broadcast use. Each participant can send a message over the bus, usually initiated by the master and answered by the addressed slave. This protocol fully follows the specifications of the Modbus TCP Protocol.

1.6 Message Format

1.6.1 Data Frame Structure

MBAP				PDU	
Transaction Identification	Protocol Identification	Length	Unit Identification	Function Code	Data
2 Bytes	2 Bytes	2 Bytes	1 Bytes	1 Bytes	n Bytes

Instructions:

- (1) Transaction identification: The sequence number of the message. Different sequence numbers are used to distinguish different messages;
- (2) Protocol identification: 0000 indicates Modbus TCP Protocol ;
- (3) Length: The length of the data after this field;
- (4) Unit identification: device address (1-255, default value:1);
- (5) Function code: Function code provided by Modbus TCP Protocol;
- (6) Data:

Modbus Request				
Read	register address	register number		
Write	register address	register number	data length	register value

Modbus Response			
Read	data length	register value	
Write	register address	register number/value	

1.6.2 Function Code

Code	Describe
0x01	Read Coils
0x05	Write Single Coil
0x0F	Write Multiple Coils
0x02	Read Discrete Inputs
0x04	Read Input Registers
0x03	Read Holding Registers
0x06	Write Single Register
0x10	Write Multiple Registers
0x14	Read File Record
0x15	Write File Record

The function codes we use are 0x03, 0x06, 0x10 and 0x15.

(1) Read Holding Registers (0x03)

Read the contents of the continuous block of Read Holding Registers from the remote device.

Request: 00 01 00 00 00 06 01 03 00 00 00 03

MBAP	Code	Reg Addr	Reg Num
00 01 00 00 00 06 01	03	00 00	00 03

Notice:

Reg Addr: Register starting address;

Reg Num: Number of registers;

Reg Val: The value of registers.

Response: 00 01 00 00 00 09 01 03 06 00 21 00 00 00 00

MBAP	Code	Data Len	Reg Val
00 01 00 00 00 09 01	03	06	00 21 00 00 00 00

(2) Write Single Register(0x06)

Write a hold register in a remote device.

Request: 00 01 00 00 00 06 01 06 00 00 00 01

MBAP	Code	Reg Addr	Reg Val
00 01 00 00 00 06 01	06	00 00	00 01

Response: 00 01 00 00 00 06 01 06 00 00 00 01

MBAP	Code	Reg Addr	Reg Val
00 01 00 00 00 06 01	06	00 00	00 01

(3) Write Multiple Registers(0x10)

Write a continuous register block in a remote device (1 to 123 registers) .

Request: 00 02 00 00 00 09 04 10 00 00 00 01 02 00 01

MBAP	Code	Reg Addr	Reg Num	Data Len	Reg Val
00 02 00 00 00 09 04 10 00 00 00 01 02 00 01	10	00 00	00 01	02	00 01

Response: 00 02 00 00 00 06 04 10 00 00 00 01

MBAP	Code	Reg Addr	Reg Num
00 02 00 00 00 06 04 10	10	00 00	00 1

(4) Write File Record(0x15)

This function code is used to perform a file record write.

Request:

MBAP	PDU
00 01 00 00 00 F8 01	The contents of the PDU are as follows

Code	1 Byte	15
Data Len	1 Byte	F5 (Range: 0x07 to 0xF5)
Subreq x, Refer Type	1 Byte	06
Subreq x, File Num	2 Bytes	00 01 (Range: 0x0000 to 0xFFFF)
Subreq x, Record Num	2 Bytes	00 00 (Register starting address)
Subreq x, Record Len	2 Bytes	00 77 (N=119)
Subreq x, Record Data	N*2 Bytes	
Subreq x+1, Refer Type		
Subreq x+1, File Num		
...		

Response: Same as request frame.

2 Register List

Reg Addr	Name	Attr	Reg Num	Len(Bytes)
0x1000	Device Address	R	1	2
0x1001	Software Version	R	1	2
0x1002	Stop Reason	R	1	2
0x1003	EVC Status	R	1	2
0x1004	CP Status	R	1	2
0x1005	CC Status	R	1	2
0x1006	Charging Port Temperature	R	1	2
0x1007	EVC Environment Temperature	R	1	2
0x1008	A Phase Voltage	R	1	2
0x1009	B Phase Voltage	R	1	2
0x100A	C Phase Voltage	R	1	2
0x100B	A Phase Current	R	1	2
0x100C	B Phase Current	R	1	2
0x100D	C Phase Current	R	1	2
0x100E	Active Power	R	1	2
0x100F	Electronic Lock Status	R	1	2
0x1010	Current Phase Sequence	R	1	2
0x1011	Max Supported Power	R	1	2
0x1012	Min Supported Power	R	1	2
0x1013	Max Supported Current	R	1	2
0x1014	Min Supported Current	R	1	2
0x1015	System Alarm Information	R	1	2
0x1016	Current Energy	R	2	4

0x1018	Total Charging Energy	R	2	4
0x101A	System Fault Information	R	2	4
0x101C	RFID Card	R	2	4
0x101E	Id Model Code	R	4	8
0x1022	Id Serial Number	R	16	32
0x3000	Work Mode	W/R	1	2
0x3001	Max Charging Current	W/R	1	2
0x3002	Max Charging Power	W/R	1	2
0x3003	Allowable Charge Time	W/R	1	2
0x3004	Allowable Charge Energy	W/R	1	2
0x3005	Time Validity	W/R	1	2
0x3006	Default Current	W/R	1	2
0x3007	OTA Status	W/R	1	2
0x3008	OTA Firmware Size	W/R	2	4
0x300A	Single/three-phase automatic switching	W/R	1	2
0x300B	EVC suspend switching minimum interval	W/R	1	2
0x4000	Electronic lock control	W	1	2
0x4001	Charging Control	W	1	2
0x4002	Phase Sequence Switching	W	1	2
0x4003	Restart	W	1	2

Note:

(1) "R": read-only, "W": write-only, "W/R": readable and writable;

(2) Write-only registers are written with 0x06, readable and writable registers are written with 0x10.

2.1 Device Address

This register describes the device address value of the EV charger in the Modbus TCP Protocol.

Address	Name	Type	Attr
0x1000	Device Address	UINT16	R

Supported values:

- Range: 1~255, and default value: 1.

2.2 Software Version

This register describes the current software version of EV charger.

Address	Name	Type	Attr
0x1001	Software Version	UINT16	R

Supported values:

- Byte1: Major version number;
- Byte0: Minor version number;

eg: If the value of this register is 0x0110, the software version is 1.16.

2.3 Stop Reason Code

This register describes the reason why EV charger stopped.

Address	Name	Type	Attr
0x1002	Stop Reason	UINT16	R

Supported values:

Details are in [Appendix 1 Stop Reason Code](#).

Notice:

When EV charger is not in the finish state, reading this register value can get the stop reason for the last charging end;

This register will be cleared after EVC restarts.

2.4 EVC Status

This register describes the EVC status information.

Address	Name	Type	Attr
0x1003	EVC Status	UINT16	R

Supported values:

- 0: idle. EVC has no faults and not connected to the car;
- 1: connect. EVC has been connected to the car and is waiting for the start command;
- 2: start. EVC has received the start command and is waiting for the car to be ready for charging;
- 3: charging. EVC is providing electric energy to the car;
- 4: pause. The charging process is suspended by the car;
- 5: finish. This charging has finished;
- 6: fault. The EVC is faulty and cannot be used for charging at this time;
- 7: reserve. (Not covered yet);
- 8: locked. In the locked state, the pile cannot perform any other operations.

2.5 CP Status

This register describes CP voltage level status.

Address	Name	Type	Attr
0x1004	CP Status	UINT16	R

Supported values:

- 0: CP fault, CP voltage is not within a normal range;
- 1: CP 12V. When EVC is not connected for charging, the CP voltage value is +12V;
- 2: CP 9V. When EVC is connected for charging, the CP voltage value is +9V;
- 3: CP 6V. When the car is ready for charging, the CP voltage value is +6V.

2.6 CC Status

This register describes the connection status of CC.

Address	Name	Type	Attr
0x1005	CC Connection Status	UINT16	R

Supported values:

- 0: Plug unconnected;
- 1: Plug connected;

2.7 Charging Port Temperature

This register describes the temperature inside the gun holder of EVC.

Address	Name	Type	Attr
---------	------	------	------

Address	Name	Type	Attr
0x1006	Charging Port Temperature	UINT16	R

Supported values:

- 0~65535, the temperature shifts upward by 50 °C, and the resolution is 0.1, unit: °C;

eg: If the value of this register is 0x0320 = 800, the actual temperature $T = 800 * 0.1 - 50 = 30$ (°C).

Notice:

The plug version of the EVC does not have this detection, and the value of this register read is invalid. For the A7300S1-E-2 model with EVC greater than or equal to V1.10 and the A022KS1-E-2 model with EVC greater than or equal to V1.20, the value of this register is fixed to 0xFFFF.

2.8 EVC Environment Temperature

This register describes the temperature around the PCB inside the EVC.

Address	Name	Type	Attr
0x1007	EVC Environment Temperature	UINT16	R

Supported values:

- 0~65535, the temperature shifts upward by 50 °C, and the resolution is 0.1, unit: °C;

2.9 A Phase Voltage

This register describes the A phase charging voltage value.

Address	Name	Type	Attr
---------	------	------	------

Address	Name	Type	Attr
0x1008	A Phase Voltage	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: V.

eg: If the value of this register is 0x08FC = 2300, the actual voltage $V = 2300 * 0.1 = 230.0$ (V).

2.10 B Phase Voltage

This register describes the B phase charging voltage value.

Address	Name	Type	Attr
0x1009	B Phase Voltage	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: V.

Consistent with the A phase voltage example.

2.11 C Phase Voltage

This register describes the C phase charging voltage value.

Address	Name	Type	Attr
0x100A	C Phase Voltage	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: V.

Consistent with the A phase voltage example.

2.12 A Phase Current

This register describes the A phase charging current value.

Address	Name	Type	Attr
0x100B	A Phase Current	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: A.

eg: If the value of this register is 0x0140 = 320, the actual current value is $320 * 0.1 = 32.0$ (A).

2.13 B Phase Current

This register describes the B phase charging current value.

Address	Name	Type	Attr
0x100C	B Phase Current	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: A.

Consistent with the A phase current example.

2.14 C Phase Current

This register describes the C phase charging current value.

Address	Name	Type	Attr
0x100D	C Phase Current	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: A.

Consistent with the A phase current example.

2.15 Active Power

This register describes the power of EVC being charged.

Address	Name	Type	Attr
0x100E	Active Power	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: KW.

eg: If the value of this register is 0x0049 = 73, the active power is $73 \times 0.1 = 7.3$ (KW).

2.16 Electronic Lock Status

This register describes the locked state of the electronics inside the EVC.

Address	Name	Type	Attr
0x100F	Electronic Lock Status	UINT16	R

Supported values:

- 0: Unlocked state;
- 1: Locked state.

2.17 Current Phase Sequence

This register describes the voltage phase of the EVC providing electric energy to the car.

Address	Name	Type	Attr
0x1010	Current Phase	UINT32	R

Sequence

Supported values:

- 0: Three-phase output;
- 1: L2 output;
- 2: L3 output;
- 3: Reserve.

Notice:

This instruction is meaningful only when the pile is connected to an external device named phase switch box.

2.18 Max Supported Power

This register describes the rated maximum working power supported by the EVC.

Address	Name	Type	Attr
0x1011	Max Supported Power	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: KW;

eg: If the value of this register is 0x0049 = 73, Max Supported Power is $73 * 0.1 = 7.3$ (KW).

Notice:

The max supported power value is determined by the model of the EVC you purchase. For example, the max supported power of 7.3 KW series EVC is 7.3 KW, the max supported power of 11 KW series EVC is 11 KW, and the max supported power of 22 KW series EVC is 22 KW . And this value is read only and cannot be

changed;

2.19 Min Supported Power

This register describes the rated Min working power supported by the EVC.

Address	Name	Type	Attr
0x1012	Min Supported Power	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: KW;

eg: If the value of this register is 0x000D = 13, the Min Supported Power is $13 * 0.1 = 1.3$ (KW).

Notice:

According to the EVC standard, the min supported power is the minimum energy required for EVC to start charging. For example, for 7.3 KW series EVC, the min supported power is 1.4 KW, for 11 KW series and 22 KW series EVC, the min supported power is 4.2 KW. This value is also read-only and cannot be changed.

2.20 Max Supported Current

This register describes the maximum rated charging current allowed for each phase of the EVC.

Address	Name	Type	Attr
0x1013	Max Supported Current	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: A;

eg: If the value of this register is $0x0140 = 320$, the Max Supported Current is $320 * 0.1 = 32.0$ (A).

Notice:

The max supported current is determined by the EVC model. For example, the maximum supported current allowed per phase of EVC with a model of 11 KW serial is 16A. This value is read-only and cannot be changed.

2.21 Min Supported Current

This register describes the minimum rated charging current allowed for each phase of the EVC.

Address	Name	Type	Attr
0x1014	Min Supported Current	UINT16	R

Supported values:

- 0~65535, the resolution is 0.1, unit: A;

eg: If the value of this register is $0x003C = 60$, the Min Supported Current is $60 * 0.1 = 6.0$ (A).

Notice:

Generally, the minimum current supported by EVC by default is 6A.

2.22 System Alarm Information

This register describes the system alarm information of EVC.

Address	Name	Type	Attr
0x1015	System Alarm Information	UINT16	R

Supported values:

Details are in [Appendix 3 Alarm Code](#).

2.23 Current Energy

This register describes the current electric energy reading of the electric meter inside the EVC.

Address	Name	Type	Attr
0x1016	Current Energy	UINT32	R

Supported values:

- 0~10000000, the resolution is 0.1, unit: kWh;

eg: If the value of this register is 0x0003F19F = 258463, the current energy is $258463 * 0.1 = 25846.3$ (kWh).

2.24 Total Charging Energy

This register describes the total electric energy currently consumed by the charging car.

Address	Name	Type	Attr
0x1018	Total Charging Energy	UINT32	R

Supported values:

- 0~10000000, the resolution is 0.1, unit: kWh;

eg: If the value of this register is 0x0003F19F = 258463, the total charging energy is $258463 * 0.1 = 25846.3$ (kWh).

2.25 System Fault Information

This register describes the system fault information of EVC.

Address	Name	Type	Attr
0x101A	System fault information	UINT32	R

Supported values:

Details are in [Appendix 2 Fault Code](#).

2.26 RFID Card

This register describes the UID of the authorized card when swiping the card to start charging.

Address	Name	Type	Attr
0x101C	RFID card	UINT32	R

Supported values:

- 0x00000000~0xFFFFFFFF.

eg: If the value of this register is 0x4F8D637E, the UID of the authorization card that starts charging this time is 4F8D637E.

2.27 Id Model Code

This register describes the Id Model Code of EVC.

Address	Name	Type	Attr
0x101E	Id Model Code	ASCII (8 Bytes)	R

Supported values:

- 0~65535.

eg: If the value of this register is 0x4130=A0, 0x3232=22, 0x4B50=KP, 0x3100=1, the type of EVC is A022KP1.

Notice:

If the last register is less than two bytes, add 0 at the end.

2.28 Id Serial Number

This register describes the Id Serial Number of EVC.

Address	Name	Type	Attr
0x1022	Id Serial Number	ASCII (32 Bytes)	R

Supported values:

- 0~65535.

eg: If the value of this register is 0x3630=60, 0x414C=AL, 0x3532=52, 0x4230=B0, 0x3341=3A, 0x4851=HQ, 0x3031=01, 0x3400=4, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, 0x0000, the serial number of EVC is 60AL52B03AHQ014.

Notice:

If the last register is less than two bytes, add 0 at the end;

If the total number of registers is less than 16, the rest of the register is replaced by 0.

2.29 Work Mode

This register describes the working mode of EVC.

Address	Name	Type	Attr
0x3000	Work mode	UINT16	W/R

Supported values:

- 0: Controlled mode. In this mode, you can start and stop charging through our company's EVC management APP "Fox Switch" or by

swiping your card;

- 1: Plug and charge mode. In this mode, charging will start immediately after plugging.
- 2: Lock mode. In this mode, no operations can be performed on the EVC.

Notice:

This register takes effect only when it is set in the charging, fault and lock state.

2.30 Max Charging Current

This register describes the maximum current allowed for this charge.

Address	Name	Type	Attr
0x3001	Max Charging Current	UINT16	W/R

Supported values:

- 0~65535, the resolution is 0.1, unit: A;

eg: If the value of this register is 0x00C8 = 200, the Max Charging Current is 200 * 0.1 = 20.0 (A).

Notice:

This register takes effect only when it is set in the charging state;

The current maximum current shall not exceed the Max Supported Current;

It will be restored to the Max Supported Current after charging;

It won't hold after a power failure.

2.31 Max Charging Power

This register describes the maximum power allowed for this charge.

Address	Name	Type	Attr
0x3002	Max Charging Power	UINT16	W/R

Supported values:

- 0~65535, the resolution is 0.1, unit: KW;

eg: If the value of this register is 0x006E = 110, the Max Charging Power is $110 * 0.1 = 11.0$ (KW).

Notice:

This register takes effect only when it is set in the charging state;

EVC will automatically convert the maximum allowable power into the allowable current of each phase;

It will be restored to the Max Supported Power after charging;

It won't hold after a power failure.

2.32 Allowable Charge Time

This register describes the maximum charging allowed for this charge .

Address	Name	Type	Attr
0x3003	Allowable Charge Time	UINT16	W/R

Supported values:

- 0~0xFFFF, unit: Minutes;

eg: If the value of this register is 0x003C = 60, EVC will stop charging after 60 minutes without other stopping conditions.

Notice:

This register is only valid in the start, charge, and pause states, and when the value is 0xffff, it means it will not take effect.

2.33 Allowable Charge Energy

This register describes the charging energy allowed this time.

Address	Name	Type	Attr
0x3004	Allowable Charge Energy	UINT16	W/R

Supported values:

- 0~0xFFFF, unit: KW·H;

eg: If the value of this register is 0x0014 = 20, EVC will stop charging when it reaches 20KW·H without other stopping conditions.

Notice:

This register is only valid in the start, charge, and pause states, andnd when the value is 0xffff,it means it will not take effect.

When the allowed charging time and the allowed charging energy take effect at the same time, EVC only needs to meet one of the conditions to stop charging. For example, the allowed charging time is 5 minutes and the allowed charging energy is 10KW·H. When charging reaches 5 minutes but the charging energy is only 2KW·H, EVC will stop charging at this time.

2.34 Time Validity

This register describes the time range how long the last command value of the EMS is valid.

Address	Name	Type	Attr
0x3005	Time Validity	UINT16	W/R

Supported values:

- Default value: 30s.

Notice:

Range: 10~60s;

If the EVC does not receive any messages within this time, it is considered offline.

2.35 Default Current

This register describes the charging current when the connection to the EMS is lost.

Address	Name	Type	Attr
0x3006	Default Current	UINT16	W/R

Supported values:

- Default value: 6A;
- 0~65535, the resolution is 0.1, unit: A.

eg: If the value of this register is 0x00C8 = 200, the default current is $200 * 0.1 = 20.0$ (A).

Notice:

Range: 6~32A.

2.36 OTA Status

This register describes the process status of the OTA upgrade of EVC.

Address	Name	Type	Attr
0x3007	OTA Status	UINT16	W/R

Supported values:

- 0: idle;

- 1: upgrading;
- 2: completed.

2.37 OTA Firmware Size

This register describes the firmware size of EVC for OTA upgrade.

Address	Name	Type	Attr
0x3008	OTA Firmware Size	UINT32	W/R

Supported values:

- OTA Firmware Size $\leq$ 128KB;

2.38 Single/three-phase automatic switching

This register describes the parameters of the single/three-phase automatic switching switch.

Function description: When this switch is turned on, the charger automatically switches between single-phase and three-phase charging according to the charging power set by the EMS.

The logic is as follows:

If $P \geq 4.2\text{KW}$, the charger operates in three-phase mode for charging;

If $P \geq 1.4\text{KW}$ and $P < 4.2\text{KW}$, the charger operates in single-phase mode for charging;

If $P < 1.4\text{KW}$, charging is paused;

Address	Name	Type	Attr
0x300A	Single/three-phase automatic switching	UINT16	W/R

Supported values:

- 0: Disabled;
- 1: Enabled;

Note:

The default value is 1, enabling the single/three-phase auto-switching function.

2.39 EVC suspend switching minimum interval

This register defines the minimum time interval required for the charger to transition from its current charging state to a suspended (or standby) state.

The time interval starts counting after the completion of the last state switch, specifying the minimum duration the charger must remain in a non-suspended state (i.e., charging or standby but not suspended) before it can enter the suspended state again.

Address	Name	Type	Attr
0x300B	EVC suspend switching minimum interval	UINT16	W/R

Supported values:

- 5 to 30, in minutes, with a default of 12 minutes;

Note:

When the minimum interval for single/three-phase auto-switching is not met, but the phase-switching conditions are satisfied, the handling logic is as follows:

If $P \geq 4.2\text{KW}$ and the charger is currently in single-phase charging, the phase will not switch, and current will be loaded in single-phase mode;

If $P \geq 1.4\text{KW}$ and $P < 4.2\text{KW}$, and the charger is currently in three-phase

charging, with the interval not met, the charger will enter a charging pause state;

If $P < 1.4\text{KW}$, and the interval is not met, the charger will also enter a charging pause state.

2.40 Electronic Lock Control

This register describes the electronic lock control instruction.

Address	Name	Type	Attr
0x4000	Electronic Lock Control	UINT16	W

Supported values:

- 0: No action;
- 1: Unlock;
- 2: Lock;

2.41 Charging Control

This register describes the start and stop charging function of EVC.

Address	Name	Type	Attr
0x4001	Charging Control	UINT16	W

Supported values:

- 0: No action;
- 1: Start charging;
- 2: Stop charging;

Note:

After executing the start and stop, you can judge whether the start and stop

are successful by reading the EVC status register.

2.42 Phase sequence Switching

This register describes the phase sequence switching instructions.

Address	Name	Type	Attr
0x4002	Phase Sequence Switching	UINT16	W

Supported values:

- 0: Three-phase power supply;
- 1: L2 power supply;
- 2: L3 power supply.

Notice:

Because it takes 1 minute from the start of charging to the stability of charging, phase cutting is allowed only after 1 minute of starting charging;

Only when the pile is connected to the phase cutting box and the pile state is charging, the phase cutting command will take effect;

Please check the current charge phase sequence register to confirm whether the phase cutting is successful.

2.43 Restart

This register describes restart EVC.

Address	Name	Type	Attr
0x4003	Restart	UINT16	W

Supported values:

- 0xA5A5: Restart.

eg: If the value of this register is 0xA5A5, EVC will restart.

3 OTA Firmware Storage Format

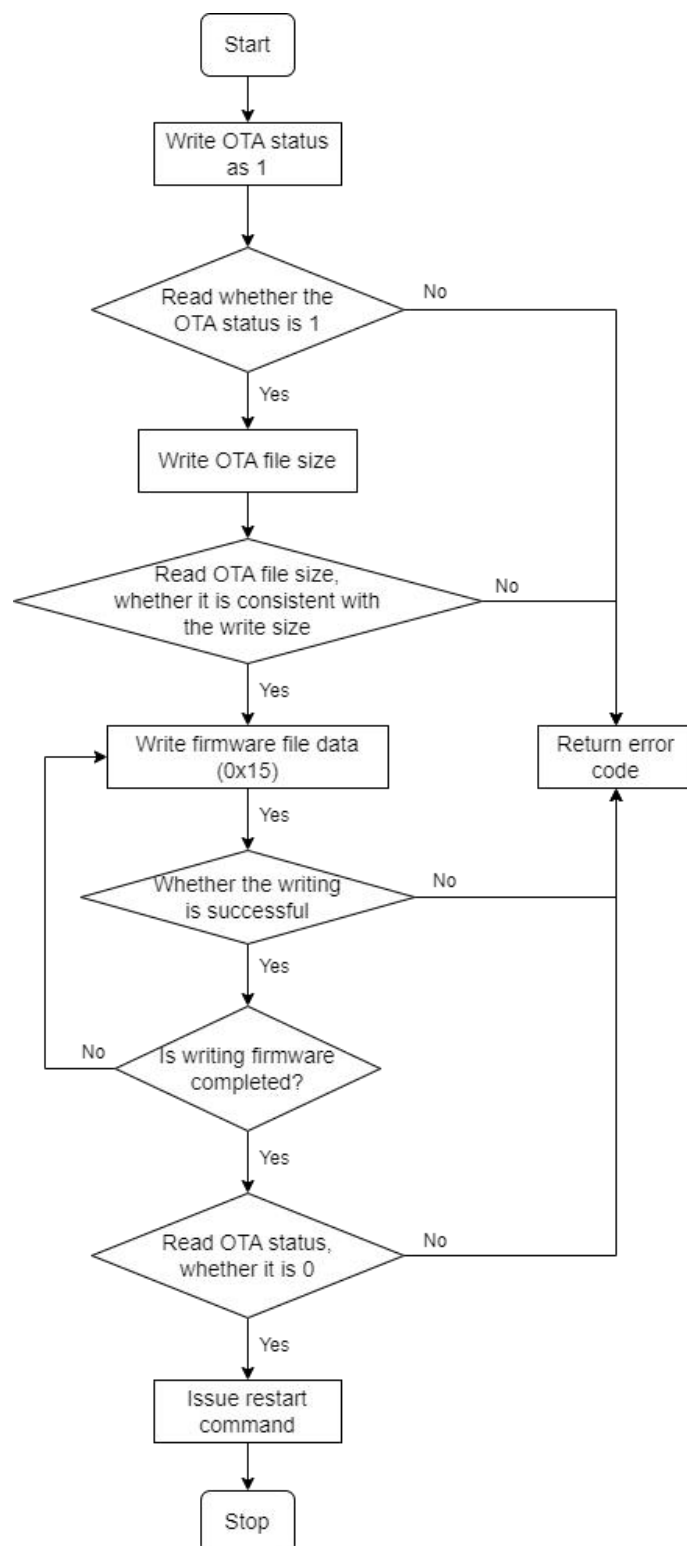
File Num	Record Num	Number of record	Data length of record (Byte)
0001	0000	0077	00EE
0001	0001	0077	00EE
0001	0002	0077	00EE
...	...	...	...
0001	N-1	0077	00EE
0001	N	T<=0x77	2T

Notice:

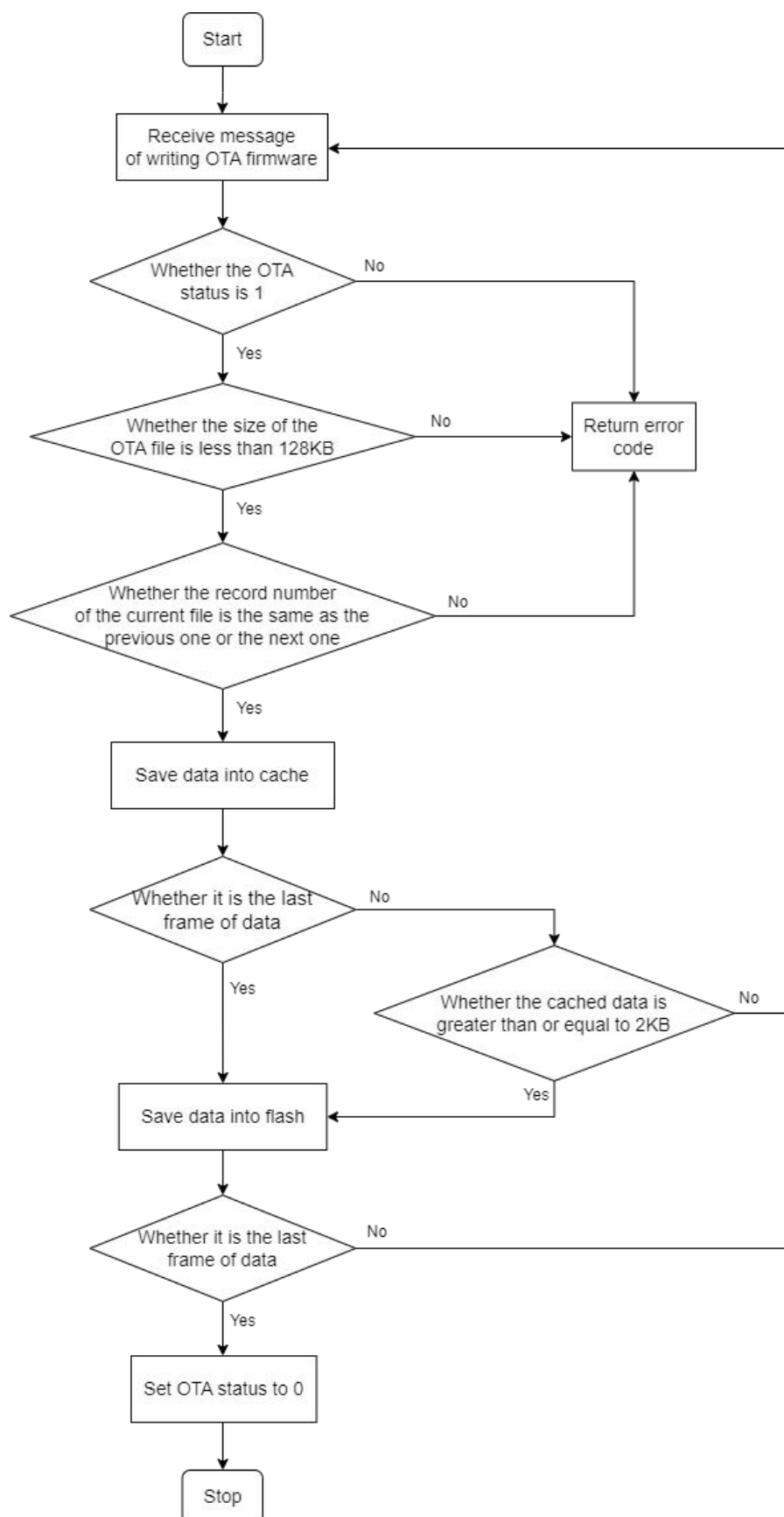
- (1) The maximum record number $N < 0x0226$;
- (2) The number of the last record $T \leq 0x77$.

4 OTA Upgrade Process

4.1 Upgrade on the host computer side



4.2 Upgrade on the EVC side



Appendix

1 Stop Reason Code

Value	Reason
0	None
1	Stopped on command
2	Timed charging completed
3	S2 timeout
4	Charging pause timeout
5	Emergency stop button pressed
6	Abnormal CP voltage
7	Abnormality in drawing a charging connector
8	Abnormal AC contactor
9	Abnormal electronic lock
10	Abnormal card reader
11	Abnormal overcurrent
12	Abnormal overvoltage
13	Abnormal undervoltage
14	Charging port over-temperature
15	Abnormal leakage current
16	N line reverse connection
17	Abnormal frequency
18	Charging stop button pressed
19	Abnormal circuit breaker
20	Phase loss
21	Abnormal PE

22	Abnormal external electric meter
23	Environment over-temperature
24	Metering chip failure
25	Access control failure
26	PBOX switching phase sequence is abnormal
27	Reach the predetermined energy

2 Fault Code

Bit	Fault name	Value
Bit0	Emergency stop	0 Normal 1 Abnormal
Bit1	Overvoltage	0 Normal 1 Abnormal
Bit2	Undervoltage	0 Normal 1 Abnormal
Bit3	Overcurrent	0 Normal 1 Abnormal
Bit4	charging port temperature	0 Normal 1 Abnormal
Bit5	PE grounding	0 Normal 1 Abnormal
Bit6	Leakage current	0 Normal 1 Abnormal
Bit7	Frequency	0 Normal 1 Abnormal
Bit8	CP	0 Normal 1 Abnormal
Bit9	Connector	0 Normal 1 Abnormal
Bit10	AC contactor	0 Normal 1 Abnormal
Bit11	Electronic lock	0 Normal 1 Abnormal
Bit12	Breaker	0 Normal 1 Abnormal
Bit13	CC	0 Normal 1 Abnormal
Bit14	External meter communication	0 Normal 1 Abnormal
Bit15	Metering chip	0 Normal 1 Abnormal
Bit16	Environment temperature	0 Normal 1 Abnormal
Bit17	Access control	0 Normal 1 Abnormal

3 Alarm Code

Bit	Alarm name	Value
Bit0	Card reader	0 Normal 1 Abnormal
Bit1	Phase cutting box	0 Normal 1 Abnormal
Bit2	Phase loss	0 Normal 1 Abnormal